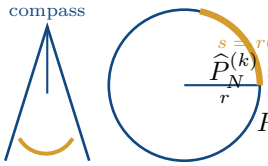
A Geometric Recasting of Classical π-Formulas

Classical formulas beside their N-Series geometric-π_g counterparts



$$\pi_g = \lim_{N\to\infty} \widehat{P}_N^{(k)}$$

$$\widehat{P}_N^{(k)} = P_{kN} + \frac{P_{kN} - P_N}{k^2 - 1}$$

$$P_N = N \sin\left(\frac{\pi}{N}\right)$$

The classical formulas below use π as the circle constant. The right-hand column rewrites the same geometry using the constructible geometric limit π_g. When the N-Series limit is established as $\pi_g = \pi$, the two columns are numerically identical; the distinction is conceptual: π_g emphasizes geometric construction, polygonal limits, and scale-cancellation structure.

Geometry	**π-form**	**π_g-form**
Circle circumference	$C = 2\pi r$	$C_g = 2\pi_g r$
Circle area	$A = \pi r^2$	$A_g = \pi_g r^2$
Arc length	$s = r\theta, \quad \theta = \frac{\alpha\pi}{180}$	$s_g = r\theta_g, \quad \theta_g = \frac{\alpha\pi_g}{180}$
Sector area	$A_{\text{sec}} = \frac{1}{2}r^2\theta = \frac{\alpha}{360}\pi r^2$	$A_{\text{sec},g} = \frac{1}{2}r^2\theta_g = \frac{\alpha}{360}\pi_g r^2$
Regular n-gon area, circumradius R	$A_n = \frac{n}{2}R^2 \sin\left(\frac{2\pi}{n}\right)$	$A_{n,g} = \frac{n}{2}R^2 \sin\left(\frac{2\pi_g}{n}\right)$
Regular n-gon side, circumradius R	$a_n = 2R\sin\left(\frac{\pi}{n}\right)$	$a_{n,g} = 2R\sin\left(\frac{\pi_g}{n}\right)$
Sphere surface area	$S = 4\pi r^2$	$S_g = 4\pi_g r^2$
Sphere volume	$V = \frac{4}{3}\pi r^3$	$V_g = \frac{4}{3}\pi_g r^3$
Cylinder volume	$V = \pi r^2 h$	$V_g = \pi_g r^2 h$
Cone volume	$V = \frac{1}{3}\pi r^2 h$	$V_g = \frac{1}{3}\pi_g r^2 h$
Torus surface area	$S = 4\pi^2 Rr$	$S_g = 4\pi_g^2 Rr$
Torus volume	$V = 2\pi^2 Rr^2$	$V_g = 2\pi_g^2 Rr^2$
Ellipse area	$A = \pi ab$	$A_g = \pi_g ab$
Ellipse perimeter approximation	$P \approx \pi\left[3(a+b) - \sqrt{(3a+b)(a+3b)}\right]$	$P_g \approx \pi_g\left[3(a+b) - \sqrt{(3a+b)(a+3b)}\right]$
Cone lateral area	$L = \pi r\ell$	$L_g = \pi_g r\ell$
Cone total area	$S = \pi r(r+\ell)$	$S_g = \pi_g r(r+\ell)$
Spherical cap area	$A_{\text{cap}} = 2\pi rh$	$A_{\text{cap},g} = 2\pi_g rh$
Spherical cap volume	$V_{\text{cap}} = \frac{\pi h^2}{3}(3r-h)$	$V_{\text{cap},g} = \frac{\pi_g h^2}{3}(3r-h)$

Classical geometry: π $\longleftrightarrow$ **Constructible N-Series geometry: π_g**

$\pi_g = \pi \implies$ every π_g-formula above reproduces the classical value,

while retaining the geometric interpretation of π_g as a polygonal, scale-accelerated, constructible circle constant.